

Problem 1 (2019F FinalExam 2). Evaluate the integral $\int_E \frac{x^2}{x^2+y^2} dV$ where E is the solid region bounded by the surface described by $z = x^2 + 2y^2$ and $z = 36 - 3x^2 - 2y^2$

Problem 2 (2019F FinalExam 5). Let C be the curve in $\mathbb{R}^2$, parametrized by the equation

$$X(t) = (\sin 2t, \sin t)$$

with $0 \leq t \leq \pi$. Denote by D the region surrounded by the curve C . Find the area of D .

Problem 3 (2019F FinalExam 6). Let S be the surface in $\mathbb{R}^3$ obtained by rotating the circle $\{(x, y, z) \in \mathbb{R}^3 : (x - 3)^2 + y^2 = 1, z = 0\}$ about the y -axis. This S , traditionally called a torus, can be parametrized by

$$X(u, v) = ((3 + \cos u) \cos v, \sin u, (3 + \cos u) \sin v)$$

with $-\pi \leq u \leq \pi$ and $-\pi \leq v \leq \pi$. Also the vector field F is given by

$$F(x, y, z) = (x + z, x + y^3, y - 2z)$$

Now, evaluate the flux integral $\int_S (\nabla \times F) \cdot N d\sigma$.

Problem 4 (2019F FinalExam 8). The vector field F on the xy -plane is given by

$$F(x, y) = \left(\frac{-y + 2}{(x - 1)^2 + (y - 2)^2}, \frac{x - 1}{(x - 1)^2 + (y - 2)^2} \right)$$

Let C_R be the circle of radius $R > 0$ centered at the point $(1, 2)$ oriented counterclockwise. Then, evaluate the work integral $\int_{C_R} F \cdot T ds$.

Problem 5 (2020F FinalExam 3). Denote by $a = (1, 0, 0)$, $b = (1, -1, 0)$, $c = (1, -1, 3)$, and $r = (x, y, z)$. Let

$$D = \{r = (x, y, z) \in \mathbb{R}^3 : 1 \leq (a \cdot r)^2 + (b \cdot r)^2 \leq 4, 0 \leq c \cdot r \leq 1\}$$

Evaluate

$$\int_D (a \cdot r)(b \cdot r)(c \cdot r) dV$$

Problem 6 (2020F FinalExam 4). The force field $F(x, y, z) = (x, -yz, xy)$ is given. Find the work done along the curve given by the intersection of the surface $y = \sin x$ ($0 \leq x \leq \pi/2$) and the plane $x = z$, from the point $(0, 0, 0)$ to the point $(\pi/2, 1, \pi/2)$.

Problem 7 (2020F FinalExam 6). Find the total outward flux of the vector field $F(x, y, z) = (x, 2y, 3z^2)$ through the boundary surface S of the solid region described by the inequalities $x^2 + y^2 \leq 1$, $y \geq 0$, $0 \leq z \leq 1$.

Problem 8 (2020F FinalExam 12). Let $A : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ and $B : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be smooth vector fields satisfying the conditions:

$$\nabla \times A = 0, \quad \nabla \cdot B = 0$$

everywhere on $\mathbb{R}^3$. Now let D be a bounded solid-region in $\mathbb{R}^3$, with a smooth boundary surface. Assume further that $A = 0$ identically in an open set containing the boundary of D . Then, show that

$$\int_D A \cdot B dV = 0$$

Problem 9 (2023F FinalExam 7). $F(x, y, z) = (yz, -xz, 1)$. Let S be the portion of surface of the paraboloid $z = 4 - x^2 - y^2$ which lies above the first octant; and let C be the closed curve $C = C_1 + C_2 + C_3$, where the curves C_1 , C_2 and C_3 are the three curves formed by intersecting S with the xy , yz and xz planes respectively (so that C is the boundary of S). Orient C so that it is traversed counterclockwise when seen from above in the first octant. Evaluate $\int_C F \cdot T ds$

Problem 10 (2023F FinalExam 8). Let F be a vector field defined by

$$F(x, y, z) = \left(\frac{z}{x^2 + z^2}, y, -\frac{x}{x^2 + z^2} \right)$$

Let C be the intersection between $x^2 + z^2 = 4$ and $x + y + z = 0$ in $\mathbb{R}^3$. Find the circulation of F over C , oriented from $(2, -2, 0)$ to $(0, -2, 2)$ (oriented counterclockwise viewed from the positive y-axis)

Problem 11 (2023F FinalExam 9). Two point charges are placed at $P_1 = (1, 0, 0)$ and $P_2 = (-1, 0, 0)$, with electric charges q_1 and q_2 , respectively. The electric field $\mathbf{F}$ on $\mathbb{R}^3 - \{P_1, P_2\}$ exerted by these two particles is given by

$$\mathbf{F}(x, y, z) = \frac{q_1}{4\pi\epsilon_0} \frac{(x-1, y, z)}{((x-1)^2 + y^2 + z^2)^{3/2}} + \frac{q_2}{4\pi\epsilon_0} \frac{(x+1, y, z)}{((x+1)^2 + y^2 + z^2)^{3/2}}$$

where ϵ_0 is the electric constant. If S is an ellipsoid $\frac{x^2}{2} + y^2 + z^2 = 1$, write the outward flux $\int_S \mathbf{F} \cdot \mathbf{N} d\sigma$ in terms of q_1 and q_2 . Justify your answer.